Assignment Question:

Victoria Road Crash Data Visualisation and Analysis

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Use of GenAi: Generative AI has been used only for paraphrasing draft texts

Word Count: 2759

Introduction:

Data Cleaning:

To ensure concurrency of data in the visuals, the three large data sets vic_road_crash_data, Person, and Vehicle had to be integrated. As there were a lot of columns, the idea was to perform targeted data cleaning/handling, and create a data frame (master_df) of the cleaned columns that can be used as a base reference to build upon the visuals by modifying the columns as I proceed.

The basic data cleaning was done in python, while R was used for the integration level cleaning.

Initial Imputation:

The key analytical columns were identified and the missing values were addressed as per the data dictionary.

This process involved rule-based imputation. For example, in the Vehicle dataset, missing **VEHICLE_BODY_STYLE** values were imputed based on the **VEHICLE_TYPE_DESC** column; like rows where **VEHICLE_TYPE_DESC** was "BICYCLE" were assigned **the VEHICLE_BODY_STYLE** of "CYCLE" (Fig. B). For other categorical columns like **CONSTRUCTION_TYPE** and **VEHICLE_MAKE**, blank entries were replaced with "Unknown" without discarding the rows (Fig. C).

Data Integration Challenge:

My initial inner_join between the Crash and Person tables returned zero rows, despite the data appearing correct. Even after standardizing the names, joins failed. The **ACCIDENT_NO** keys which appeared identical had hidden whitespace and inconsistent capitalisation, thus both keys were made to be *toupper(str_trim())* before any join, which finally resolved the 0-row issue (Fig E).

While creating the master_df , the number of rows returned 0 again. The filter *[SEX %in% c("M", "F")]* was removing all data. Checking the raw cleaned_person.csv , revealed that column had “Male” and “Female” instead of M & F which was corrected according to the data dictionary.

My assumption of accident_date and time to be ymd_hms() assumption which had to be corrected using lubridate::mdy_hms().

Strategic Pivot:

For **VEHICLE_TYPE_DESC** had extremely long, inconsistent labels so instead despite cleaning it earlier, so I used the vic_road_crash to create a a new vehicle_category column (motorcycle, heavy vehicle, passengervehicle etc) (Fig F).

For the **VEHICLE_MAKE** column as well there were many unique entries such as synonyms ("MITSUB" vs. "MITSUBISHI"); inconsistent formatting ("B M W" vs. "BMW"); Multiple “Unknown ” Types. A simple top 10 crashes would have incorrectly treated these as separate makes.

To solve this I : first identified the **top 30 most common raw entries** and using this wrote a case_when() to manually consolidate the known synonyms (for e.g MITSUB to MITSUBISHI; B M W to BMW). The rest were grouped in a single category (OTHER) using fct_lump_n() (Fig G)

Load the vehicle dataset

Subtask:

Load the `vehicle.csv` file into a pandas DataFrame.

```
df_vehicles = pd.read_csv('/content/vehicle.csv')
display(df_vehicles.head())
display(df_vehicles.info())
display(df_vehicles.isnull().sum())
```

	ACCIDENT_NO	VEHICLE_ID	VEHICLE_YEAR_MANUF	VEHICLE_DCA_CODE	INITIAL_DIRECTION	ROAD_SURFACE_TYPE	ROAD_SURFACE_TYPE_DESC	REG_STATE	VEHICLE_BODY_STYLE	VEHICLE_MAKE	...	TRAILER_TYPE	VEHICLE_COLOUR_1	VEHICLE_COLO
0	T20120001479	A	NaN	1.0	N	9.0	Not known	NaN	NaN	NaN	...	H	ZZ	
1	T20120002094	C	1997.0	8.0	W	1.0	Paved	V	SEDAN	MITSUB	...	H	RED	
2	T20120002390	A	2001.0	1.0	NE	3.0	Gravel	V	SEDAN	HOLDEN	...	H	GRY	
3	T20120004794	A	NaN	1.0	E	1.0	Paved	NaN	NaN	UNKN	...	H	ZZ	
4	T20120010082	B	2005.0	2.0	S	1.0	Paved	V	SEDAN	TOYOTA	...	H	SIL	

5 rows × 37 columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 340713 entries, 0 to 340712
Data columns (total 37 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   ACCIDENT_NO           340713 non-null  object
 1   VEHICLE_ID            340713 non-null  object
 2   VEHICLE_YEAR_MANUF    333775 non-null  float64
 3   VEHICLE_DCA_CODE      339867 non-null  float64
 4   INITIAL_DIRECTION     340712 non-null  object
 5   ROAD_SURFACE_TYPE     340712 non-null  float64
 6   ROAD_SURFACE_TYPE_DESC 340712 non-null  object
 7   REG_STATE             319732 non-null  object
```

Fig A: Load the CSV

Impute missing values in `VEHICLE_BODY_STYLE` using VEHICLE_TYPE_DESC

: Impute missing `VEHICLE_BODY_STYLE` values with 'CYCLE' specifically for rows where `VEHICLE_TYPE_DESC` is 'BICYCLE'

```
df_vehicles.loc[df_vehicles['VEHICLE_TYPE_DESC'] == 'Bicycle', 'VEHICLE_BODY_STYLE'] = df_vehicles.loc[df_vehicles['VEHICLE_TYPE_DESC'] == 'Bicycle', 'VEHICLE_BODY_STYLE'].fillna('CYCLE')
display(df_vehicles.loc[df_vehicles['VEHICLE_TYPE_DESC'] == 'BICYCLE', 'VEHICLE_BODY_STYLE'].isnull().sum())
```

```
np.int64(0)
```

Fig B: Imputing VEHICLE_BODY_STYLE Missing values

Impute missing values in `CONSTRUCTION_TYPE`

Replace blank values in the `CONSTRUCTION_TYPE` column with "unknown".

```
df_vehicles['CONSTRUCTION_TYPE'].fillna('unknown', inplace=True)
print(df_vehicles['CONSTRUCTION_TYPE'].isnull().sum())
```

```
0
/tmp/ipython-input-739623451.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using a
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always b

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)

  df_vehicles['CONSTRUCTION_TYPE'].fillna('unknown', inplace=True)
```

Fig C: Imputing CONSTRUCTION_TYPE missing values

Verify imputation

```
print(df_vehicles[['REG_STATE', 'VEHICLE_BODY_STYLE', 'VEHICLE_MAKE', 'CONSTRUCTION_TYPE']].isnull().sum())
```

```
REG_STATE              0
VEHICLE_BODY_STYLE     7502
VEHICLE_MAKE           0
CONSTRUCTION_TYPE      67111
dtype: int64
```

Fig D: Checking Null Values

```
clean_crash <- crash_data %>%
  mutate(ACCIDENT_NO = toupper(str_trim(ACCIDENT_NO))) %>%

  select(ACCIDENT_NO, LGA_NAME, LATITUDE, LONGITUDE) %>%
  #Fix the coordinates
  filter(
    !is.na(LATITUDE) & !is.na(LONGITUDE) &
    LATITUDE > -39.2 & LATITUDE < -33.9 &
    LONGITUDE > 140.9 & LONGITUDE < 150.0
  )
```

Fig E: toupper(str_trim()

```
# 2. Create our NEW, CLEAN 'vehicle_category'
mutate(
  vehicle_category = case_when(
    MOTORCYCLE > 0 ~ "Motorcycle",
    PASSENGERVEHICLE > 0 ~ "Passenger Vehicle",
    HEAVYVEHICLE > 0 ~ "Heavy Vehicle",
    TRUE ~ "Other" # This will catch bicycles, etc.
  )
) %>%
```

Fig F: new vehicle_category column

```
  # 3. Clean VEHICLE_MAKE
  mutate(
    VEHICLE_MAKE_CLEAN = toupper(str_trim(VEHICLE_MAKE)),
    VEHICLE_MAKE_CLEAN = case_when(
      str_detect(VEHICLE_MAKE_CLEAN, "MERC") ~ "MERCEDES BENZ",
      str_detect(VEHICLE_MAKE_CLEAN, "TOYO") ~ "TOYOTA",
      str_detect(VEHICLE_MAKE_CLEAN, "HOLD") ~ "HOLDEN",
      str_detect(VEHICLE_MAKE_CLEAN, "FORD") ~ "FORD",
      str_detect(VEHICLE_MAKE_CLEAN, "NISS") ~ "NISSAN",
      str_detect(VEHICLE_MAKE_CLEAN, "HYUN") ~ "HYUNDAI",
      str_detect(VEHICLE_MAKE_CLEAN, "MITSU") ~ "MITSUBISHI",
      str_detect(VEHICLE_MAKE_CLEAN, "MAZDA") ~ "MAZDA",
      str_detect(VEHICLE_MAKE_CLEAN, "SUBARU") ~ "SUBARU",
      str_detect(VEHICLE_MAKE_CLEAN, "HONDA") ~ "HONDA",
      str_detect(VEHICLE_MAKE_CLEAN, "VOLKS") ~ "VOLKSWAGEN",
      TRUE ~ VEHICLE_MAKE_CLEAN
    ),
    make_clean = fct_lump_n(VEHICLE_MAKE_CLEAN, n = 10)
  ) %>%
```

Fig G: Recode and Lump

2. Visualisation-Analysis

Visualisation 1: Time series visualisation to examine trends in road crashes

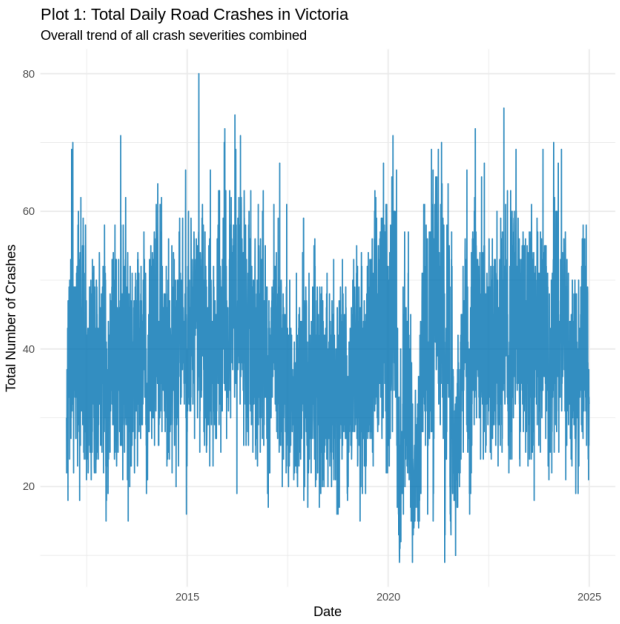


Fig 1.1: Total Daily Road Crash

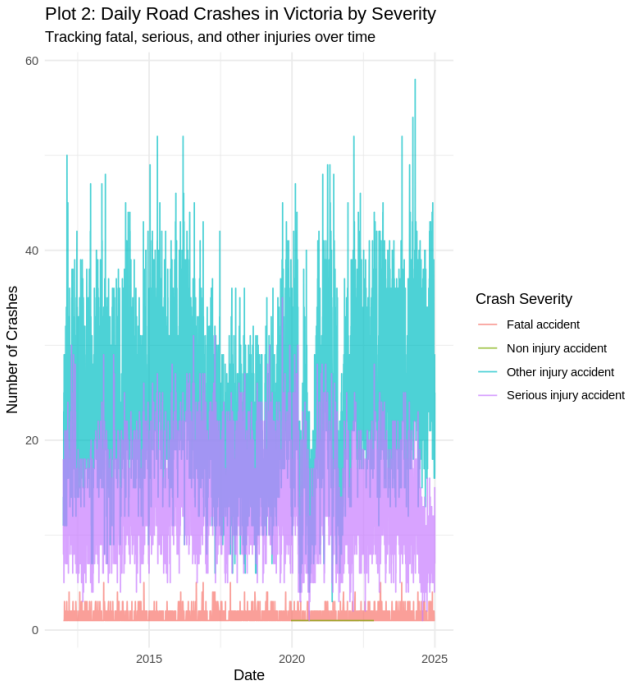


Fig 1.2: Daily Crashes by Severity

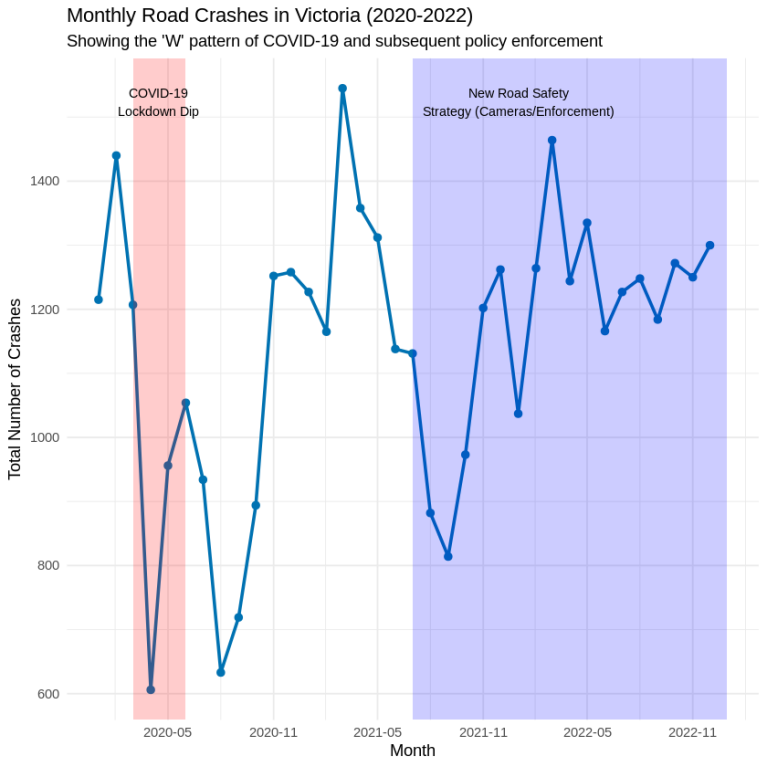


Fig 1.3: Monthly Crashes time-series (Jan 2020- Dec 2022)

A preliminary plot of the entire 12-year daily dataset was created to get an overview of the trends (Fig 1.1, 1.2). The chart clearly revealed two distinct dips in total crashes, forming a "W" shape.

I chose the monthly aggregation and a filtered 3-year timeframe (2020-2022). This selection was made to specifically investigate the "W" pattern, a significant anomaly observed in the data.

The first, sharp decline beginning in March 2020 is a clear exposure anomaly. It directly correlates with the implementation of Victoria's first COVID-19 lockdown measures. This reduction in crashes is not due to improved road safety but to a drastic, non-voluntary reduction in traffic volume.

The second dip which forms the latter half of the W (in 2021-2022) is more analytically significant. While traffic volume began to return, crashes did not return to pre-pandemic levels. This pattern strongly suggests a new, external factor was at play.

Research confirmed this hypothesis. This second dip aligns directly with the launch and aggressive enforcement of the Victorian Road Safety Strategy 2021-2030 to halve road deaths and progressively reduce serious injuries by 2030(Transport Accident Commission, 2021).

This strategy was launched in 2021 with the specific goal of halving road deaths. Critically, it was implemented with a series of "short-term action plans" (Government of Victoria, 2022) that included:

1. **Strengthened Enforcement:** A massive rollout of new **mobile phone and seatbelt detection cameras** using AI (Transport Accident Commission, 2021).
2. **Targeted Police Operations:** Victoria Police's own strategy for 2021-2024 focused on "intentional high risk driver behaviour," "speeding," and "distraction" (Transport Accident Commission, 2021).
3. **Increased Fines:** Millions of fines were issued by these new cameras, specifically targeting speeding and mobile phone use (Government of Victoria, 2023).

The conclusion is that the 2020-2022 trend occurred due to pandemic exposure-driven anomaly (first dip), while second dip was the policy-driven anomaly (technology-based enforcement).

Visualisation 2: Geographic distribution of road crashes across different regions in Victoria.

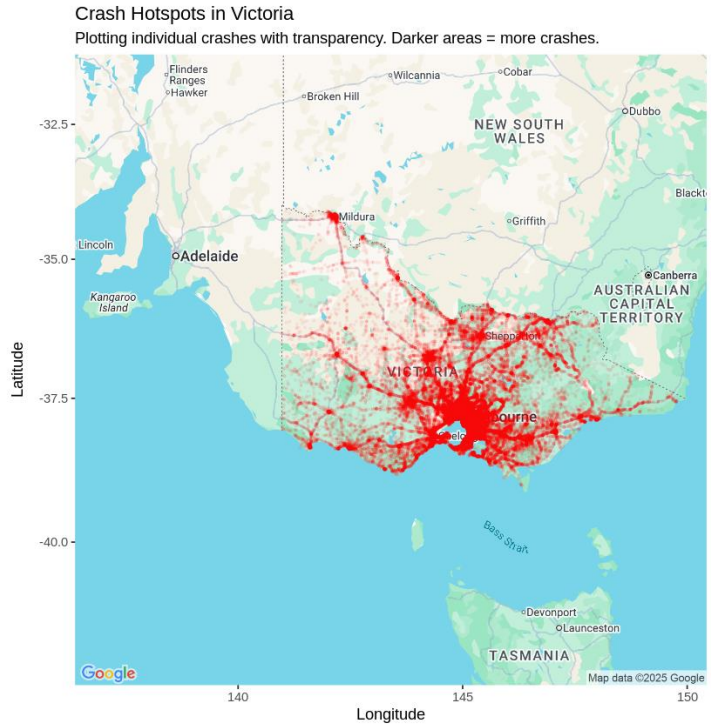


Fig 2.1 Overview of Victoria's Crash hotspots

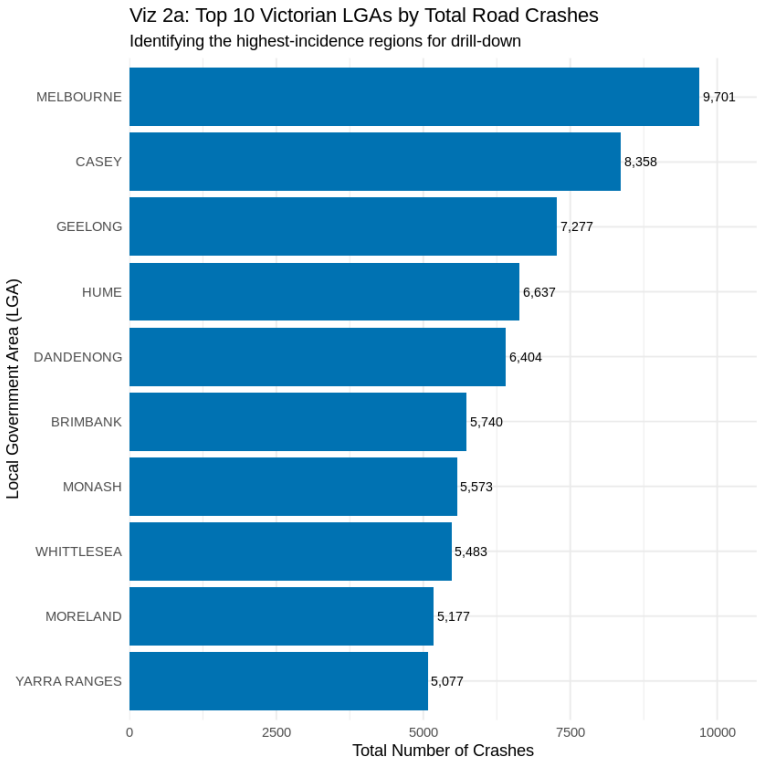


Fig 2.2 Top 10 Victorian LGAs by Total Road Crashes

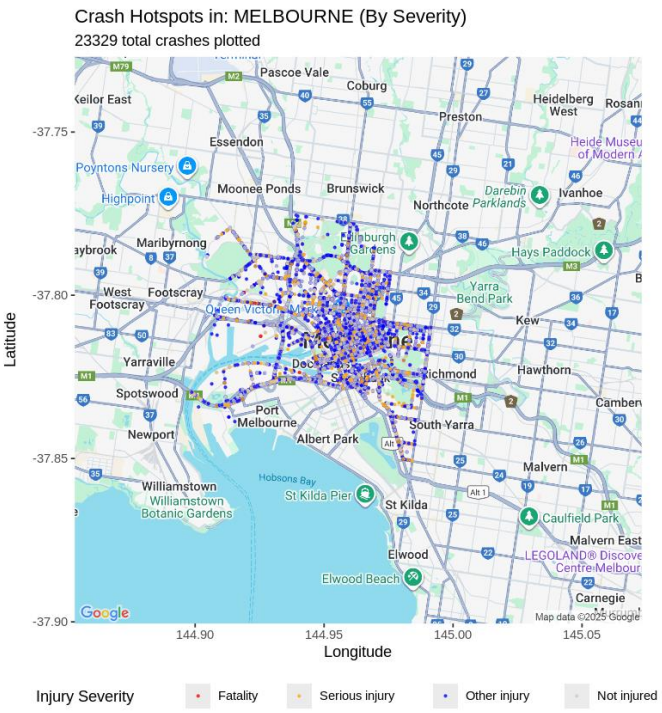


Fig 2.3 Melbourne Crash Hotspots

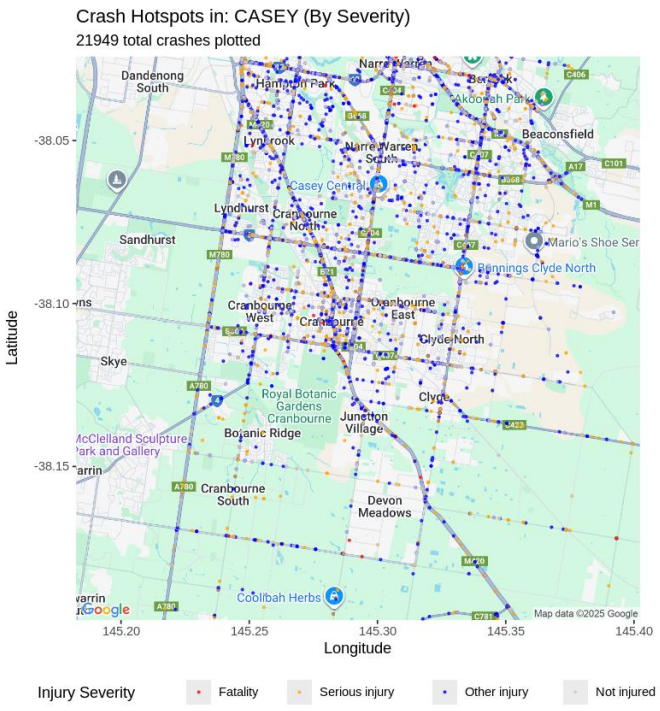


Fig 2.4 Casey Crash Hotspots

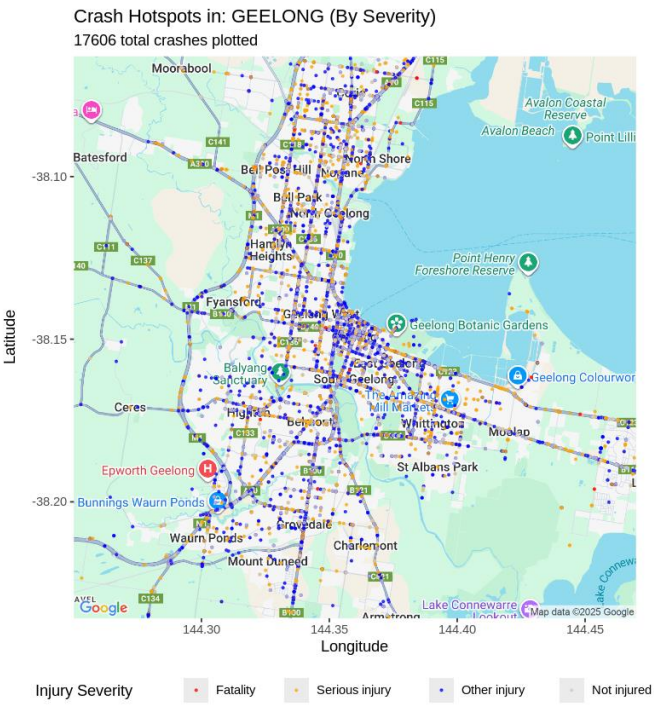


Fig 2.5 Geelong Crash Hotspots

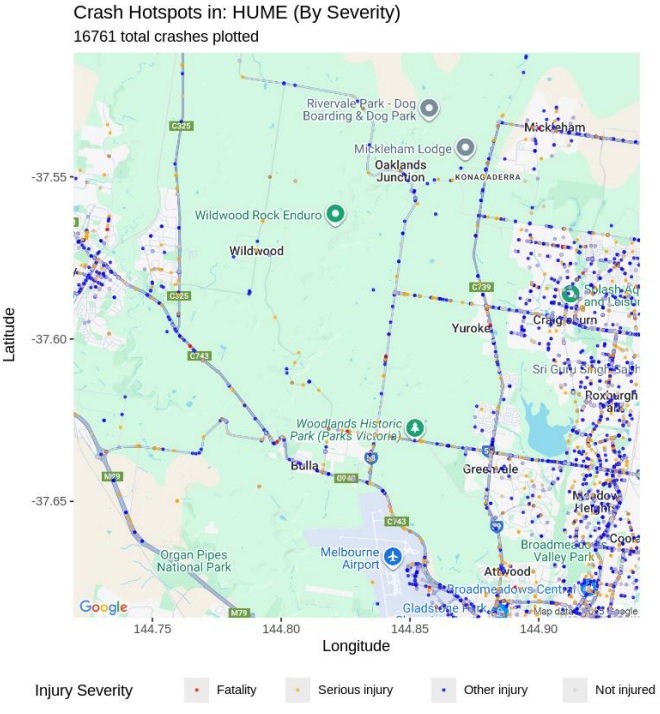


Fig 2.6 Hume Crash Hotspots

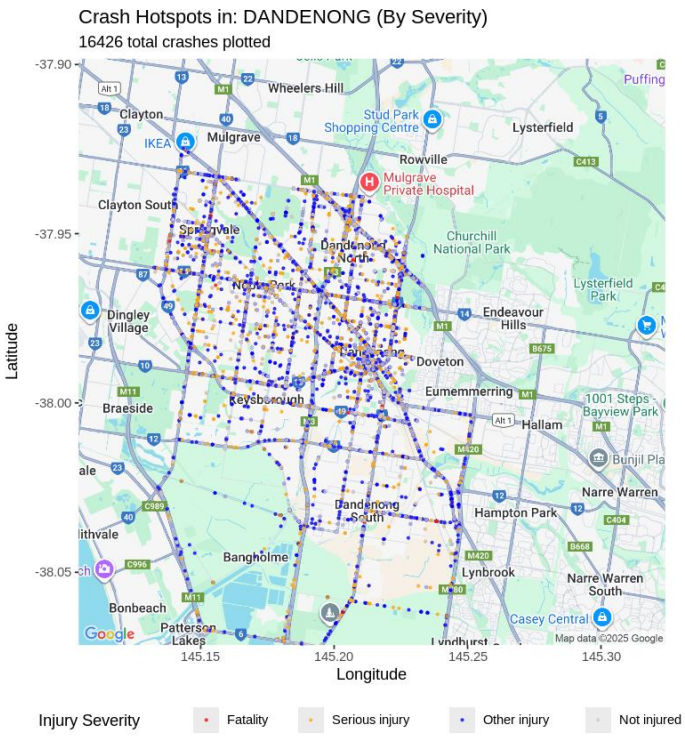


Fig 2.7 Dandenong Crash Hotspots

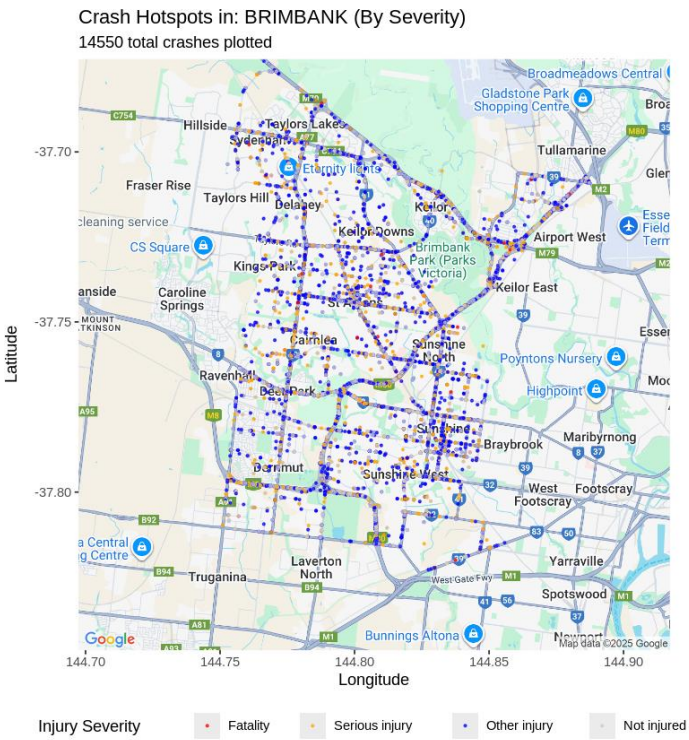


Fig 2.8 Brimbank Crash Hotspots

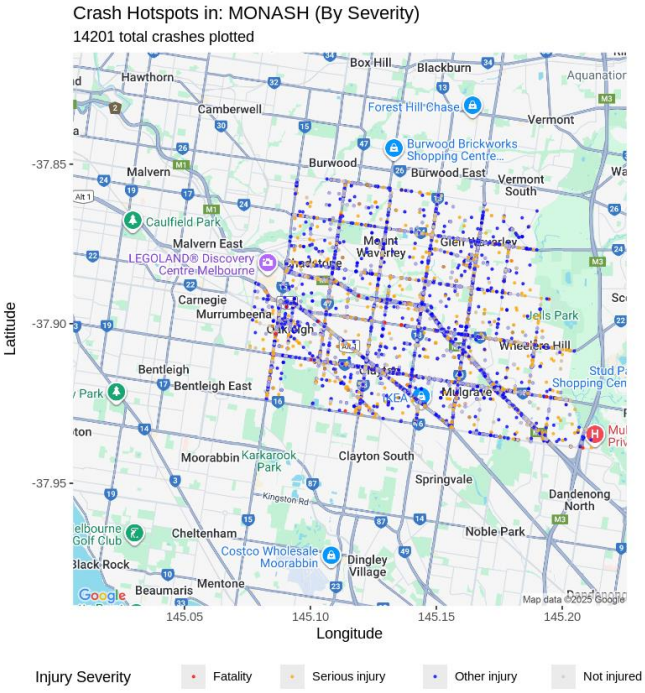


Fig 2.9 Monash Crash Hotspots

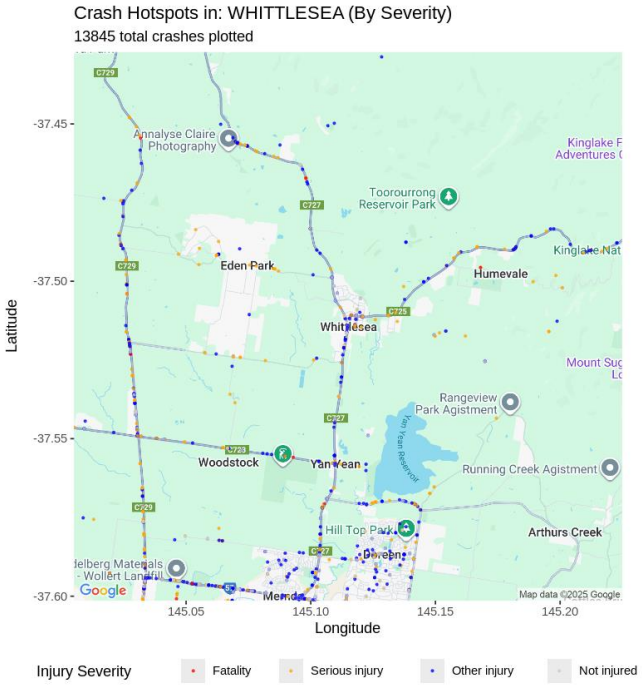


Fig 2.10 Whittlesea Crash Hotspots

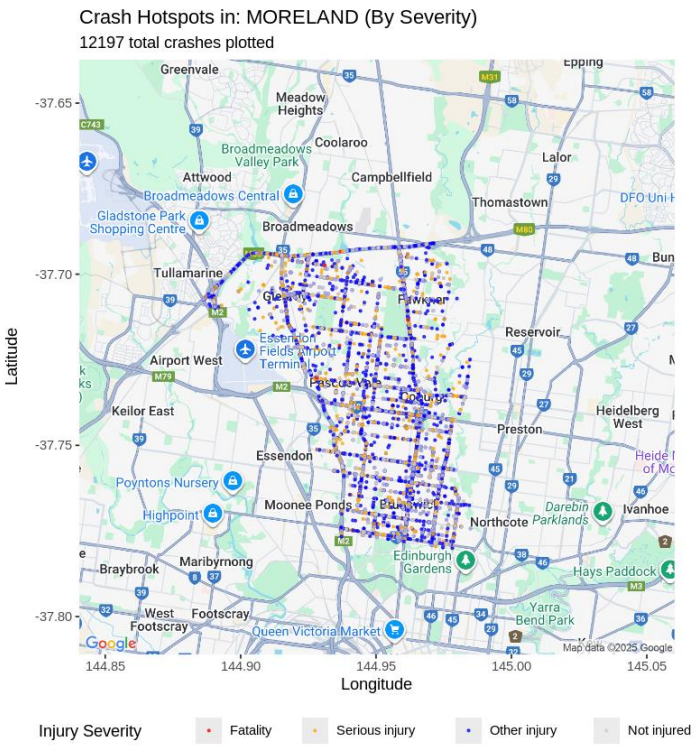


Fig 2.11 Moreland Crash Hotspots

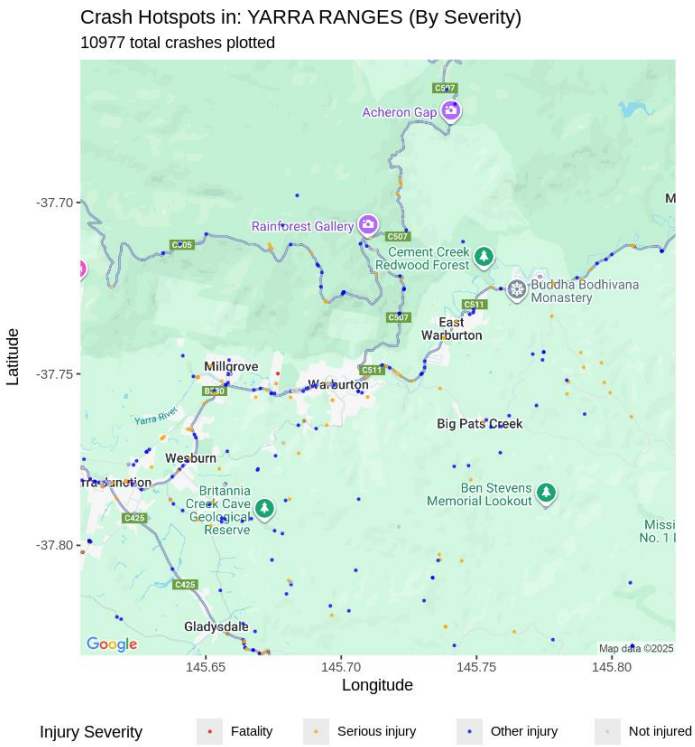


Fig 2.12 Yarra Ranges Crash Hotspots

A single state-wide map was created to understand the crash distributions. The idea was to extract the top 10 crashes lga's and visualise them on the map. The inital analysis revealed that over 50% of all incidents are concentrated in the Melbourne metro area, which would render a state-wide map with a large crash blob, that simply confirms the obvious (more people = more crashes).

The Top 10 LGA's gives a more targeted approach for visualising the crashes along with each crash point colored by injury severity (Fatality, Serious Injury, etc.).

In-depth Analysis and Interpretation: Viz 2a served as our "selection tool," identifying the 10 LGAs with the highest absolute crash volume. The subsequent drill-down (Viz 2b) was the main analysis. It revealed that "hotspots" are not monolithic; rather, they fall into **four distinct crash signatures**, each with different underlying risk factors.

1. **The "CBD Choke Point" (Melbourne, Geelong):** The maps for Melbourne and Geelong show crashes heavily concentrated in a central "blob" (Fig 2.3, 2.5). In Melbourne, this blob aligns with the M1 (Monash Freeway) as it merges and conflicts with the dense inner-city grid. In Geelong, the pattern is similar, with crashes clustered at the major intersections connecting its eastern and western halves. This represents a **high-volume, high-conflict** risk driven by traffic density and complex merging at the intersections
2. **The "Suburban Corridor" (Casey, Brimbank, Monash, Dandenong):** These large, sprawling suburban LGAs show a different pattern. The crashes are not in one blob like a single CBD but are distributed along the entire road network hierarchy. There are distinct clusters along major motorways (M-routes, like the M1), primary arterial roads (A, B, and C-routes), and the smaller roads that feed them. This is a **commuter and freight** pattern, where the risk is spread across the main corridors people use to travel to and from the city (Fig 2.4, 2.7, 2.8, 2.9).

3. **The "Urban Grid" (Moreland):** The map for Moreland, an inner-urban area, shows risk concentrated on its own city routes (like Sydney Road). This is an **urban-density** pattern. The risk is not from freeways but from a complex, low-speed environment with a high mix of pedestrians, trams, buses, and one of the city's highest rates of cyclist traffic.
4. **The "Rural/Recreational Risk" (Hume, Yarra Ranges, Whittlesea):** These "fringe" or “peri-urban” LGAs revealed a unique pattern. They are a mix of suburban sprawl and vast, rural/forested/mountainous parkland. The maps show crashes (including a high number of **red "Fatality" dots**) clustered off the main suburban grid. They are concentrated along winding, high-speed rural roads (like Route 58 in Hume) and in "green" zones like parks (Toorourrong Reservoir, Fig 2.10) and recreational lookouts (Ben Steves Memorial Lookout, Fig 2.12).

   The risk is driven by:

- **High Speeds + Complex Geometry:** Winding, high-speed roads with poor visibility.
- **Unfamiliar Drivers:** Drivers are often tourists who are unfamiliar with the road's hazards.
- **Different Vehicle Mix:** Trails such as Wildwood Rock Enduro Park (Fig 2.6)is where there’s a higher concentration of **motorcyclists** and off-road vehicles.
- **Environmental Hazards:** These "green" areas mean a much higher risk of wildlife on the road, especially at dusk.

Visualisation 3: Severity of crashes and varying severity levels across relevant factors

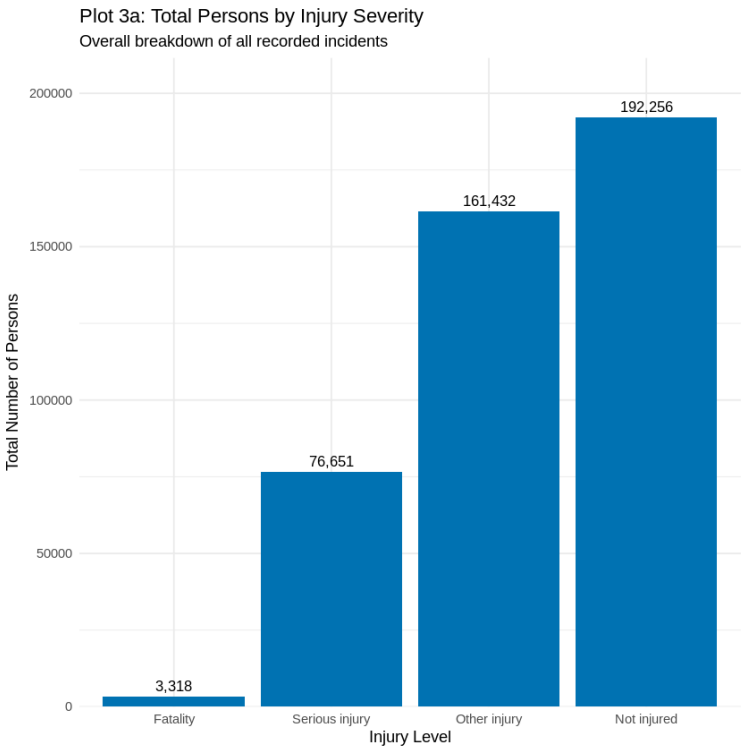


Fig 3.1: Total Persons by Injury Severity

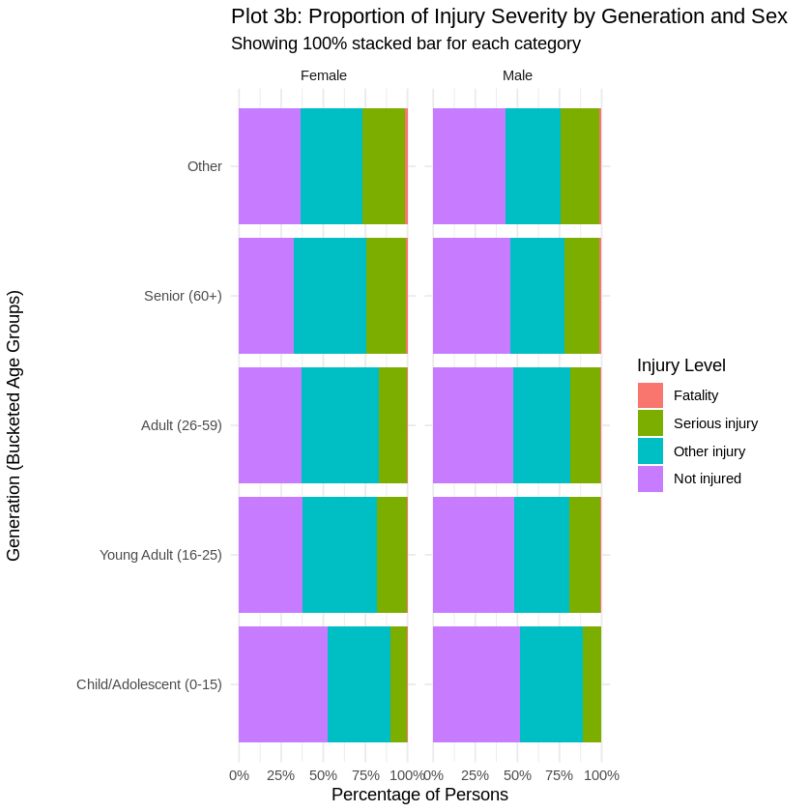


Fig 3.2 Severity by Generation and Gender

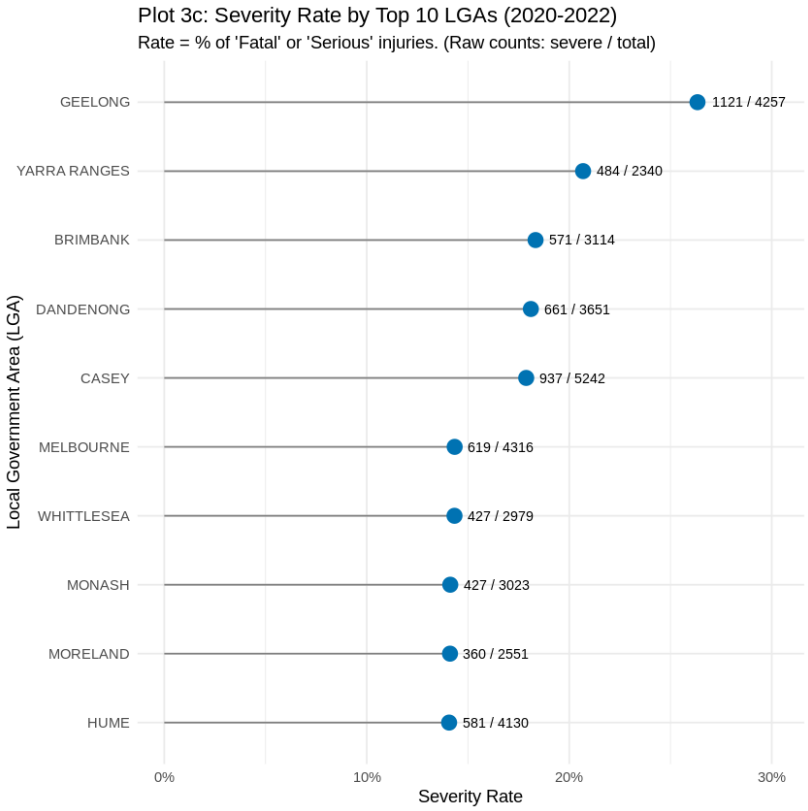


Fig 3.3 Descending Severity Rate by Top 10 LGA's

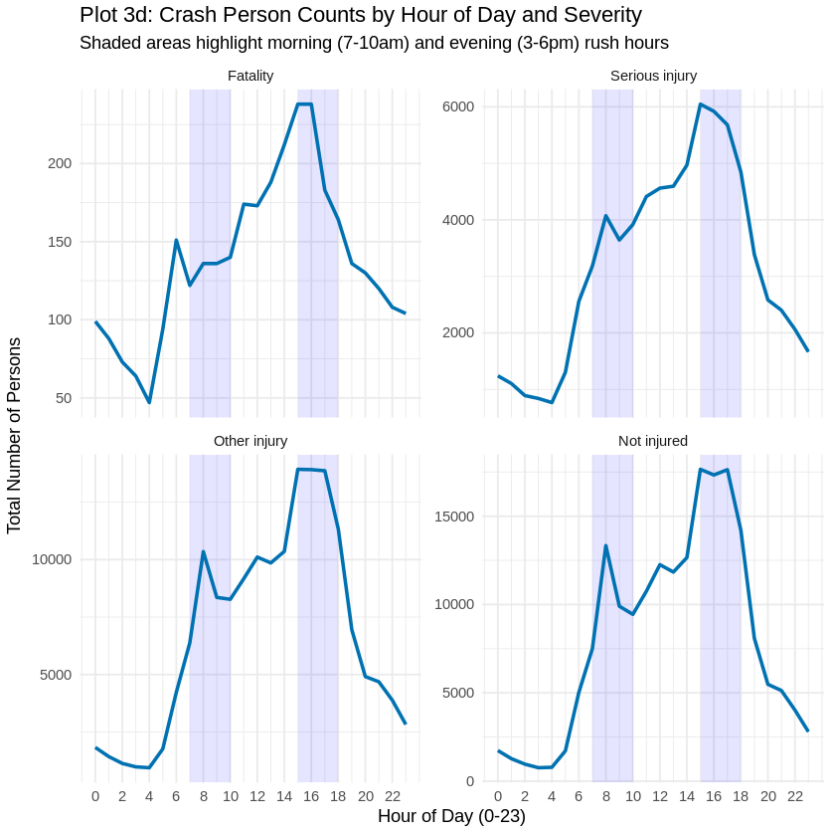


Fig 3.4 Crash counts by hour and Severity

Fig 3.1 represents a baselines scale to visualise the severity outcomes. This was chosen over pie-chart as the fatality category (3318) would have been a near-invisible strip. The bar-chart shows that "**Not injured"** (192,256) and "**Other Injury"** (161,432) are most common outcomes.

Fig 3.2 aimed to test if demographic factors influence injury severity. Due to non-uniform AGE_GROUP column a new sorted generation column (Young Adult, Senior) was created to enable analytically valid age buckets.

This plot proves that crash risk is not uniform across a persons' life. With Child/Adolescent group (0-15) forming the largest proportion of "Not Injured" outcomes, meaning the child-harness and safety has been taken care of by their guardians. However, a small proportion of male child/adult injuries were fatal as well.

Although broadly the Adult (26-59) constituting the majority of drivers and Young Adult (16-25) show similar severity patterns, close observation reveals that the young-adults are prone to serious injuries more than the adults. For the seniors, especially males the other/serious/fatal injuries become more prominent than any other age group.

Both the "young-adults" and "Seniors" are prone to more serious/fatal injuries slices visibly larger than adult group. The trend of age driven fatality is consistent across Male and Female facets indicating a prominent age-based factor than gender.

- **Young Adults**, this strongly suggests: a combination of inexperience and higher risk-taking
- For **Seniors**, this suggests a vulnerability: a crash of any given force is more likely to result in a severe injury.

Recommendations: A **bifurcated safety campaign**.

1. **For Young Adults:** Behavioural Intervention through "shock" ads targeting speed and phone; continued strict law-enforcement.
2. **For Seniors:** Promoting "safer driver" refresher courses and encouraging the adoption of newer vehicles with modern safety features.

Fig 3.3 involves a new metric "Severity Rate" (% of Fatal/Serious injuries), to separate risk from volume. Raw counts represent the severe(fatal/serious) injuries over total injuries.

There is a "Volume vs Risk" observable difference. Although Melbourne has the highest number of crashes (Fig 2.2), it's only 6th in terms of its Severity Rate (Fig 3.3). Whereas Geelong, having 3rd highest crash volume but highest severity rate (1121/ 4257). Meaning, person involved in a crash in Geelong was proportionately more likely to be killed/seriously injured than any other LGAs.

A city's safety strategy would be flawed if it only focused on the LGAs with the highest *volume* of crashes.

Recommendations:

1. **For High-Volume LGAs (Melbourne, Casey):** Traffic-optimization, intersection redesign (noted in Fig 2.3)
2. **For High-Risk LGAs (Geelong):** Determine specific risk-enhancement factors (e.g., road design, or specific "Rural/Recreational Risk" (Viz 2) and implement targeted interventions.

Fig 3.4 represents a 24-hour trend of crash severity. The peak morning and evening rush periods are highlighted to analyse the 2 Peaks prevalent across the 4 crash categories. Observation reveals a trend: An increase in crash-injuries during the morning rush and then a decrease during noon, and another steep climb around the evening rush with a sharp decline during night. This trend of 2 maxima peaks is irrespective of the injury severity. In-depth trend reveals 2 distinct crash signatures:

1. The "Not injured" and "Other injury" facets follow a similar pattern, gradually increase during dawn, with sharp peaks at the morning, plateaus during evening rush hours. This is a high-volume, low-severity problem driven by low-speed collisions. Thus traffic-flow and congestions must be managed.
2. While “Fatality” and “Serious Injury” have a sharp increase during dawn and another sharp increase to the evening rush hour, then gradually declining over-night. This elevated late-night/early-morning pattern is due to high-risk behaviours prevalent speed, alcohol/drug impairment, and fatigue. Thus, a high-visibility enforcement and PSA’s is a must prevent strategy.

Visualisation 4: Varying Crash severity across vehicle factors

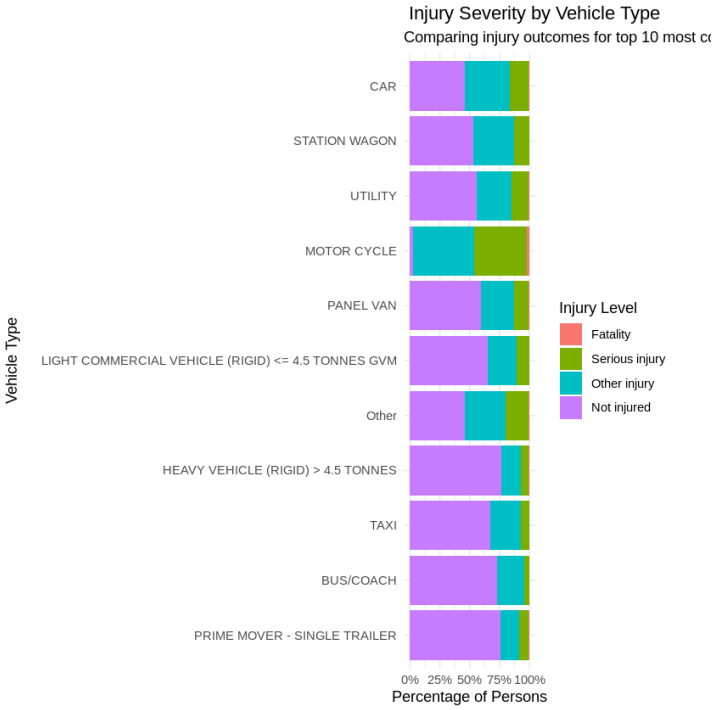


Fig 4.1: Injury Severity by Vehicle Type (descending order of crashes)

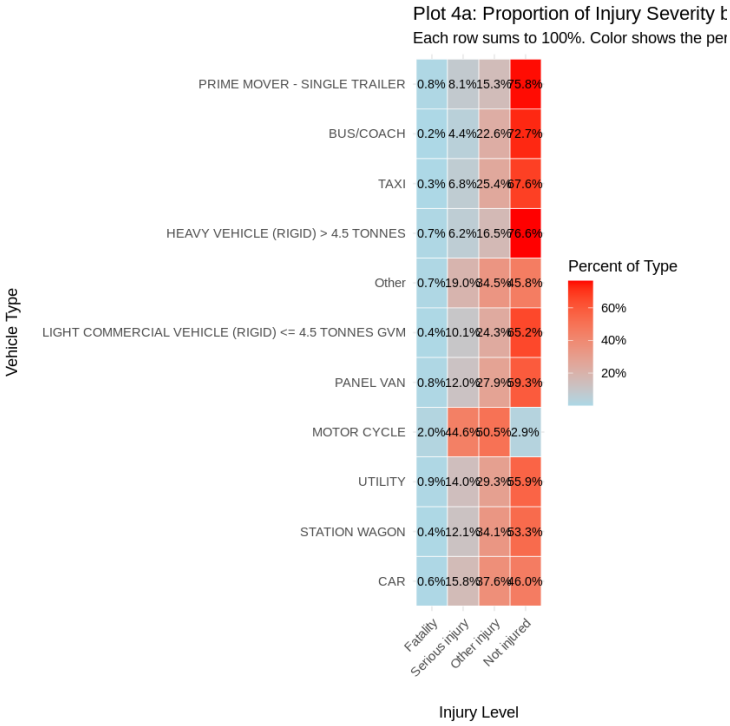


Fig 4.1.a: Injury Severity with percentages of injury level

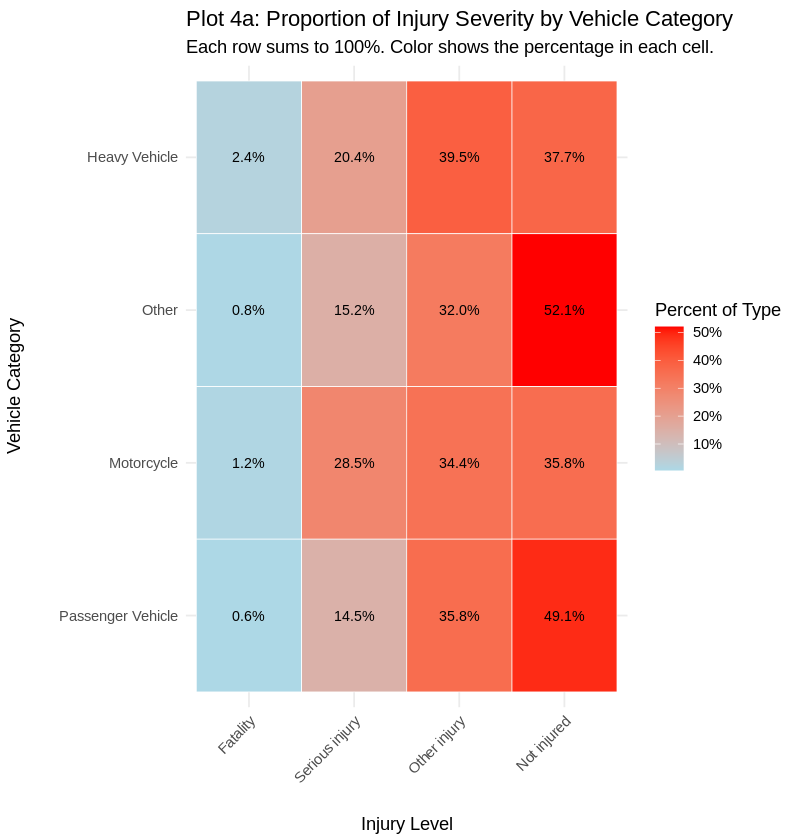


Fig 4.1.b Simplified Injury severity by Vehicle Category (% = Injury share for that category))

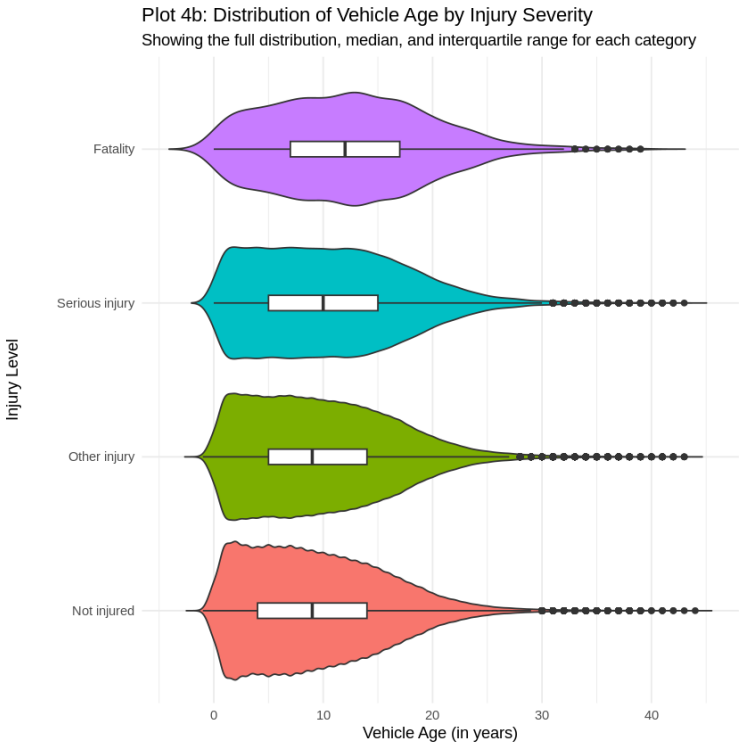


Fig 4.2: Vehicle Age and Injury level

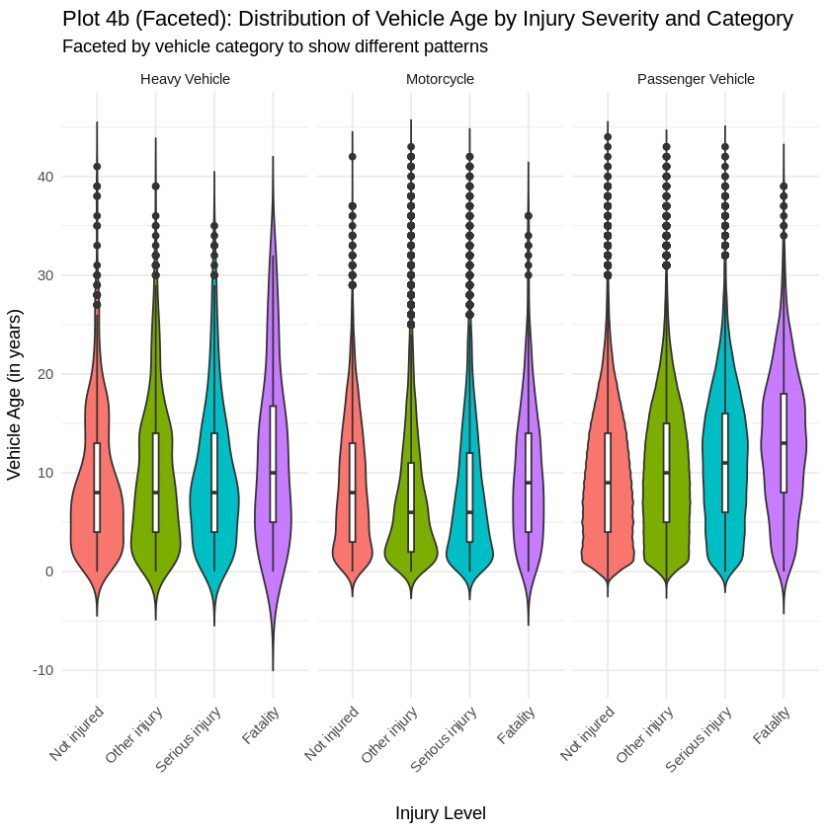


Fig 4.3 Age distribution by category and their severity

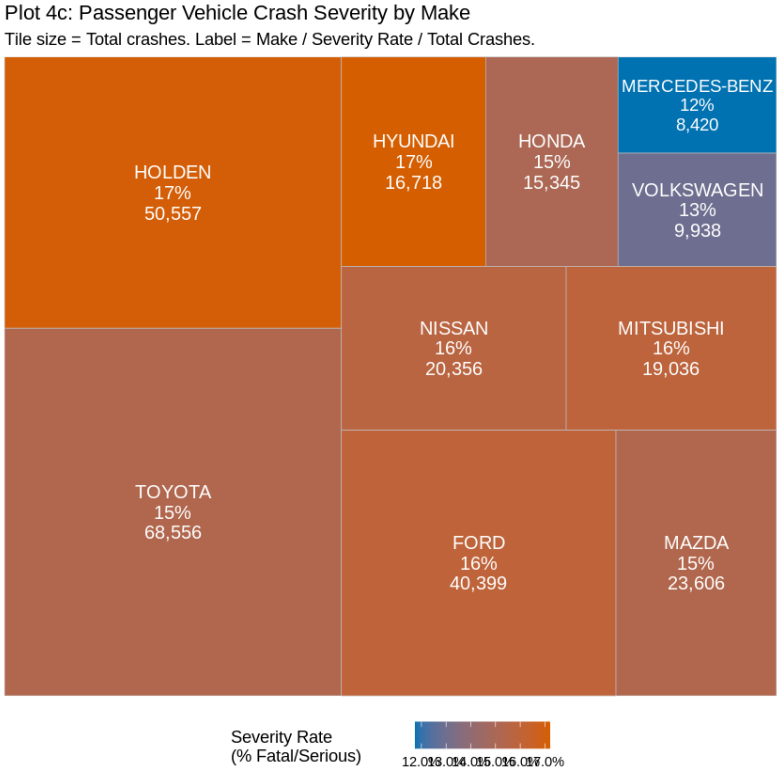


Fig 4.4 Severity By Make

Fig 4.1 represent the injury severity by vehicles in a descending order of number of crashes. Notably, Car, Station-wagon and Utility were involved in most crashes but the occupants were unharmed (unharmed car: 46%, station-wagon: 53%, Utility: 55%, Fig 4.1.a). However, motorcycle-riders suffered levels of injuries.

To visualise it as separate categories, vehicle_category high-level column was created using vic_road_crash.

Fig 4.1.b reveals:

1. For “Passenger Vehicle” the 2 common outcomes are not-injured (~50%) , other-injury(~36%) and same for “heavy-vehicles” ~80% of crashes are in these categories, with 2.0% being fatal. Problem is of volume and congestion with intervention focus on traffic flow, collision prevention.
2. For Motorcycle, however the vast majority of crashes, result in other/serious injury or fatality. ~65% of crashes in motorcycle will have serious impacts. Risk is of “**occupancy vulnerability”**. Every crash is a severe event. Interventions should include "look twice for bikes" campaigns aimed at car drivers, mandatory high-spec helmet/protective gear advocacy.

Fig 4.2 reveals the average-age vs severity pattern, which as-is reveals that older vehicles have more fatal crashes than the newer ones, due to lack of safety features, vehicle’s life-time degradation etc. With the “fatal” violins’ median at around 15 yrs and the belly (records) being concentrated around 10+years of age.

Fig 4.3 faceted data revealed risk-signatures of vehicle-category and Age

1. **Heavy Vehicles**: "Not injured" and "Other injury" outcomes, the median vehicle age is low, concentrated at <10 years, the belly also represents the bulk of the working fleet. However, "Fatality" crashes, the pattern inverts, and the entire distribution shifts upwards. The "belly" of the violin is elongated, and the median) is clearly centered around 15+ years. While newer trucks are involved in more total crashes, incidents involving older heavy vehicles are disproportionately fatal. The older trucks often lack modern safety features like Electronic Stability Control (ESC) or Automatic Emergency Braking (AEB), thus pose a distinct public risk. Policy intervention is necessary that either mandate safety retrofitting for such vehicles or create a "phase-out" scheme.
2. **Motorcycles**: The vehicle age is not the significant factor in crash severity. All four violins, from "Not injured" to "Fatality," are similarly shaped and heavily concentrated on newer bikes. proves that the risk is not in the age of the bike but is inherent to riding. A crash on a brand-new motorcycle is as severe as a crash on an old one, reinforcing the "occupant vulnerability” and its intervention efforts.
3. **Passenger-Vehicles**: While the median age for "Fatality" crashes is slightly higher than for "Not injured," the distributions show significant overlap. This suggests that while newer cars with 5-star ANCAP ratings and modern safety features are beneficial, the age-based "safety gap" is far less distinct than the ones observed in the heavy vehicle fleet. Interventions would involve continuous consumer awareness on link between 5-Star ANCAP-rated vehicles and safety outcomes.

Fig 4.4 shows the "Severity Rates" (% of Fatal/Serious injuries):

- **The "Baseline" Group:** Most high-volume Japanese and Korean-makes cluster at a similar, lower baseline of risk. This includes **Mazda (15%)**, **Mitsubishi (16%)**, **Nissan (16%)**, **Toyota (15%)**. It also shows that the largest brands by crash-volume: Toyota are not brands with highest risk. This seperates commonly-found from proportional-severity
- **The "Outlier" Group:** We identified two distinct high-risk groups.
    1. **Holden (17%)** and **Hyundai (17%)**, show a slightly elevated severity rate compared to the baseline.
    2. The luxury **Mercedes-Benz (12%)**, although has the lowest crash volumes in this group, they are associated with the high-proportional rate of severe injuries.

Recommendations: It proves that focusing a safety strategy *only* on the highest-volume brands (like Toyota) would be a mistake, as data shows high-end cars are at equal risk as well. Thus, messaging should focus on the fact that their higher severity rate is not a reflection of the car's safety, but likely a direct consequence of high-risk behaviours such as speed.

Visualisation 5: Crash Severity Rate by Light and Road Conditions

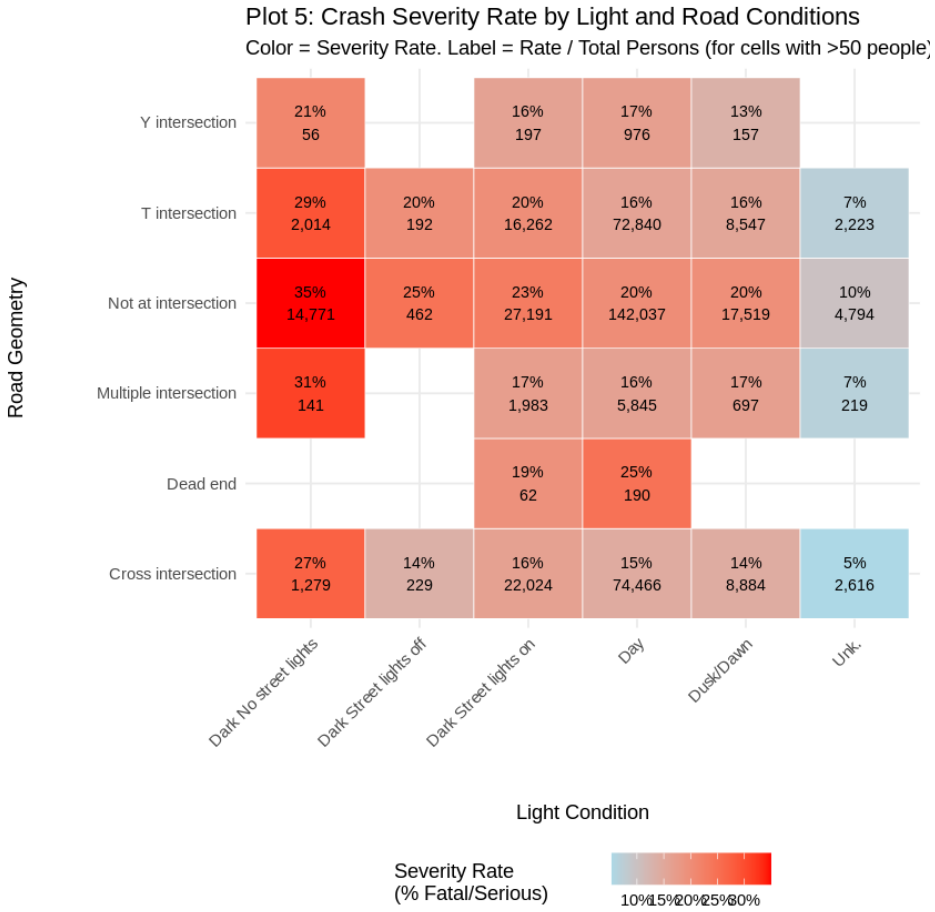


Fig 5: Crash Severity by Light/Road Conditions

Fig 5 shows that the risk associated with driving at night is as dependent on infrastructure, as on darkness itself.

The heatmap reveals a stark contrast:

- **Baseline:** During the "Day," intersections have a low Severity Rate. Observing the "Dark - Street lights on," these rates are almost identical. This proves that when streetlights are present, driving at night at an intersection is statistically no more dangerous than in broad daylight.
- **"Lethal" Interaction:** The pattern *explodes* in the "Dark - No street lights" column. For e.g The Severity Rate for: multiple-intersection almost doubles from 16% to 31%, and for a T-Junction jumps from 16% to 29%.
- The danger is not "driving at night" or "intersections" as separate factors; it is the risk of an **unlit intersection**.

Recommendations:

A targeted, state-wide program to **install streetlights at unlit intersections and Cross-roads** would directly address a known, high-risk pattern and would have a measurable impact on reducing fatal and serious injuries.

Ethical Framework and Conclusion

1. Honesty through Context

- **Principle:** It is unethical to present a statistic that, while technically true, misleads the audience.
- I ensured statistics were not misleading. For example, Fig 3.3 for LGA severity was enhanced with raw counts (e.g., "1121 / 4257") to provide full transparency, preventing a high percentage from a small sample size from looking disproportionately dangerous.

2. Correcting for Inherent Data-Bias

- **Principle:** If the raw data contains biases plotting it means an unethical replication of that bias.
- I ensured visuals were fair. The raw AGE_GROUP column was non-uniform and biased; re-engineered it into a logical generation column to ensure Fig 3.2 was based on fair, uniform buckets.

3. Data Aggregation for Privacy

- **Principle:** Even in a public dataset, data must be handled in a way that protects individual privacy. The LATITUDE and LONGITUDE columns are highly sensitive.
- When using sensitive LATITUDE/LONGITUDE data, I used high-level aggregations (heatmaps, density plots) and grouped data by LGA, showing regional patterns without revealing the exact location of any single, tragic event.

Conclusion:

This analysis of Victorian road data revealed that crash risk is not uniform but follows distinct, actionable patterns. Time-series analysis linked crash rates to both public health events (COVID) and policy enforcement. Geographically, classified hotspots into four unique signatures, from "CBD Choke Points" to "Rural/Recreational Risk."

Severity is observed to be disproportionately high for young/senior drivers, during late-night hours, and for vulnerable road users like motorcyclists. Risk is *multiplicative* an unlit intersection is far more dangerous than just "dark" and "intersection" as separate factors. This report provides a clear, evidence-based framework for targeted, multi-pronged safety interventions.

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